Volume 1 — Machine Overview & Safety

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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Chapter 1 — About the Machine

The Mazak **Vertical Quality Center VQC-20/40** is a vertical machining center originally delivered with a **Mazatrol M-2** conversational CNC control (the 41434WB schematic set is titled M-1 because it is the original 10/1983 drawing set; the NC on SN 060231 was upgraded to M-2).

Item	Detail
Model	Mazak VQC-20/40
Serial number	060231
Original control	Mazatrol M-2 (Meldas-based)
Axes	X, Y, Z — analog velocity-command servo drives (Mitsubishi TRA type)
Servo motors	Mitsubishi HD81-12S class, tachometer velocity feedback
Position feedback	**Tamagawa resolvers** (documented: RT-5XA-11, RT-5XA-L1; TS2014N noted per axis in pin authority)
Spindle drive	Mitsubishi **FR-SX** with analog speed reference
Spindle gearbox	Hi/Lo hydraulic shift (SOL-12 / SOL-13, PRS-10 / PRS-12 confirm)
ATC	Hydraulic tool changer + BCD-coded magazine (PRS-21…25)
Tooling	Per Mazak Tooling Manual M00T002 (BT/CAT 40-50 family)
Incoming power	460 VAC 3-phase; onboard variable-voltage transformer tapped for 460 V, stepping to the 230 VAC internal bus. **No external transformer or phase converter is required.**
Control power	24 VDC bus from HR-11F-24 supply (P24/G24 distribution)

Chapter 2 — The Conversion at a Glance

The retrofit replaces **only the Mazatrol M-2 NC unit** (the *NUMERICAL CONTROL UNIT* block on schematic sheet 4143075312). Everything below stays:

Retained: X/Y/Z servo amplifiers, servo motors and tachometers, Tamagawa resolvers, FR-SX spindle drive, hydraulic power unit and ATC, magazine, solenoids and prox switches, 24 VDC control bus, cabinet terminal strips, transformer and power distribution.

Replaced / Added:

Old (Mazatrol M-2)	New (LinuxCNC/Mesa)
Mazatrol NC unit & CRT/keyboard	LinuxCNC control PC + monitor
NC-to-machine interface	**Mesa 7i97T** Ethernet analog servo controller (static IP 192.168.1.121, `hm2_eth`)
Resolver-to-NC feedback path	**Mesa 7i49** resolver-to-digital interface (plain, 5 kHz excitation baseline)
Mazatrol PLC I/O rack	**Mesa 7i84U** remote field I/O on smart-serial, near the green breakout PCB
Mazatrol conversational programming	Standard G-code (plus optional WHB04B USB pendant)

The original PLC ladder logic (manual YM2V39L, 385 catalogued signals) defines the interlock behavior that the LinuxCNC/HAL configuration must reproduce.

Chapter 3 — General Safety Rules

1. **Only trained personnel** may operate, wire, or service this machine.
2. **Read Volume 3 before energizing anything.** During bring-up, most I/O assignments are unverified plans.
3. Never bypass, jumper, or software-defeat a safety interlock — door switch, overtravel limit, or E-stop.
4. Keep guards and doors closed during spindle rotation and program execution.
5. Know where every E-stop is before you need one.
6. No loose clothing, jewelry, or unsecured long hair near the spindle or ATC.
7. Verify workholding and tool retention (correct pull studs per M00T002) before cycle start.
8. The ATC and magazine move with hydraulic force — stay clear of the tool-change envelope whenever hydraulics are pressurized.

Chapter 4 — The Safety Chain (E-stop Philosophy)

Rule 1 of this retrofit: a **hardware** safety chain must remove hazardous power. LinuxCNC and HAL logic **supplement** the chain — they are never the only thing standing between an E-stop button and moving iron.

- The E-stop string (buttons, door interlocks, overtravel finals, servo contactor) is wired in hardware. Pressing E-stop drops the servo/hydraulic contactors **directly**, regardless of the PC.

- The Mesa 7i97T monitors the chain on TB5.10 (`estop-ext`) so LinuxCNC *knows* the chain opened and can abort motion cleanly — monitoring, not actuation.
- Original chain wiring must be **traced and recorded before any control rewiring** (see Volume 3, Ch. 9).
- After any change to the chain, perform the full E-stop functional test in Volume 3 §9 before production use.

Chapter 5 — Lockout / Tagout & Electrical Safety

- **460 VAC 3-phase** enters the cabinet; the transformer, drive bus, and spindle drive remain lethal after the door opens. Lock out at the wall disconnect and verify dead before work.
- Drive DC bus capacitors hold charge after power-off — wait and measure before touching drive terminals.
- The 24 VDC field bus (HR-11F-24) powers solenoids and relays. A "safe" 24 V circuit can still fire a hydraulic valve — treat it with respect during ATC work.
- One person, one lock. Test-before-touch with a meter you have proven on a known-live source.

Chapter 6 — Retrofit-Specific Hazards

Hazard	Why it exists	Defense
Axis runaway	A reversed analog command polarity makes the servo run away at full speed the moment it is enabled.	Bring-up procedure enables one axis at a time, drives inhibited first, zero-command verified, low gain (Vol 3 §8).
Resolver mis-wiring	Original Meldas/TRA wiring may drive the resolver "backwards" (excitation into stator). The 7i49 expects the opposite.	Ohmmeter the winding pairs **before power**; never trust original wire names (Vol 4 §3).
Dual excitation	If any legacy circuit still drives the resolver windings while the 7i49 is connected, signals and hardware can be damaged.	The 7i49 must be the **sole** excitation source — verified before energizing.
Unverified I/O states	NO/NC and active-high/low assumptions in the HAL files are placeholders. A wrong assumption can fire a solenoid on boot.	Every output stays disconnected until its normal state is measured (PLAN_VERIFY discipline, Vol 4 §10).
Over-current outputs	Solenoid and contactor coils (some 100 VAC) exceed Mesa output ratings.	Interposing relays RLY-1…7 with flyback/RC suppression (Vol 4 §§4, 9).
ATC motion faults	ATC forward/reverse outputs energized without proven interlocks can crash the changer.	ATC outputs are last in bring-up order, dry-run only, no tool load (Vol 3 §8, step 8).

End of Volume 1. Continue to Volume 2 — Operation Manual.